Sheet (1)

(1) The formula for converting a temperature from Celsius (C) to Fahrenheit (F) is given below:

$$F = \frac{9}{5}C + 32$$

Write a Python program that prompts the user to enter a temperature in Celsius, converts it to Fahrenheit using the given formula, and displays the resulting temperature.

(2) The cross-sectional area and volume of a cylinder with a given diameter and length are calculated using the formulas given below:

$$area = \frac{\pi}{4}.D^2$$

$$Volume = area \times Length$$

Write a Python program that prompts the user to enter the diameter and length of a cylinder, then calculates and displays its cross-sectional area and volume.

(3) Write a Python program that prompts the user to enter a length in feet, converts the length to meters, and displays the result. Assume that one foot is 0.305 meters.

(4) Write a Python program that converts pounds into kilograms. The program prompts the user to enter a value in pounds, converts it to kilograms, and displays the result. One pound is 0.454 kilograms.

(5) Write a Python program that prompts the user to enter the minutes (e.g., 1 billion, 1×10^9), and displays the number of years and days for the minutes. For simplicity, assume a year has 365 days.

(6) Write a Python program that calculates the energy needed to heat water from an initial temperature (T_0) to a final temperature (T_f). Your program should prompt the user to enter the amount of water in kilograms and the initial and final temperatures of the water. The formula to compute the energy is

$$Q = M(T_f - T_0) \times 4187$$

where M is the weight of water in kilograms, temperatures are in degrees Celsius, and energy Q is measured in joules.

(7) Average acceleration is defined as the change of velocity divided by the time taken to make the change, as shown in the following formula:

$$a = \frac{v_1 - v_0}{t}$$

Write a Python program that prompts the user to enter the starting velocity v_0 in $meters/second$, the ending velocity v_1 in $meters/second$, and the time span t in $seconds$, and displays the average acceleration.

(8) Given an airplane's acceleration a and take-off speed v, you can compute the minimum runway length needed for an airplane to take off using the following formula:

$$length = \frac{v^2}{2a}$$

Write a Python program that prompts the user to enter v in $meters/second$ and the acceleration a in $meters/second^2$ and displays the minimum runway length.

(9) The distance between two points in the $x - y$ coordinate plane is calculated using the following formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Write a Python program that prompts the user to enter the coordinates of two points, calculates the distance between them using the given formula, and displays the result.

(10) Write a Python program that prompts the user to enter the coordinates of the three vertices, $(x_1\ ,\ y_1)$, $(x_2\ ,\ y_2)$, and $(x_3\ ,\ y_3)$ of a triangle. The program should then calculate and display the area of the triangle. Use the given formulas.

$$side = \sqrt{(x_n - x_{n-1})^2 + (y_n - y_{n-1})^2}$$

$$s = \frac{1}{2}\ (side_1 + side_2 + side_3)$$

$$area = \sqrt{s(s - side_1)(s - side_2)(s - side_3)}$$

(11) Write a Python program that prompts the user to enter the side length of a hexagon and displays its area. The formula for computing the area of a hexagon is:

$$area = \frac{3\sqrt{3}}{2}s^2$$

where s is the length of a side.

(12) According to population statistics in Egypt at the end of 2025, there is:
There are 226.625 births per hour
There are 70.583 deaths per hour
Assume that the population at the end of 2025 is 118,058,996 and one year has 365 days. Write a Python program to estimate and display the population for each of the next five years.